KOUSHIL VARMA KAKARLAPUDI

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SUMMARY

Systems Engineer with expertise in DevOps, cloud infrastructure, and distributed systems. Certified Oracle Cloud DevOps Professional with hands-on experience in containerization, CI/CD automation, and monitoring solutions. Full-stack developer proficient in backend and frontend technologies. Strong foundation in Cpp ,OOP and systems design.

EDUCATION

KL University — B.Tech Computer Science & Engineering (Honors) — Specialization: Software Modelling & DevOps
CGPA: 8.5/10.0 — Most Recent Semester: 9.57/10.0 — 3rd Year Complete, 4th Year Upcoming Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, Go, Linux, Bash
Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, CI/CD, Terraform, Nginx
Backend: Node.js, Express.js, Spring Boot, REST APIs.
Databases: PostgreSQL, MySQL, MongoDB, Redis
Frontend: React.js, Next.js, TypeScript, Tailwind CSS
Tools: Git, GitHub, Jira, Prometheus, Grafana, System Design, OOP, Agile

PROJECTS

Real-Time Distributed System with DevOps Pipeline Mar 2026 – Present
Kubernetes, Docker, Jenkins, AWS — GitHub

- Architected distributed real-time system with microservices on Kubernetes handling concurrent requests.
- Designed CI/CD pipeline using Jenkins with automated Docker builds and AWS deployment strategies.
- Implemented monitoring using Prometheus and Grafana for system health tracking and incident detection.
- Applied resilience patterns: circuit breakers, retry logic, and graceful degradation.

System Monitoring & Observability Platform Apr 2026 – Present
Prometheus, Grafana, ELK Stack, Kubernetes, Docker — GitHub

- Built observability platform integrating metrics collection, centralized logging (ELK), and distributed tracing.
- Designed Grafana dashboards visualizing latency percentiles, error rates, and request throughput.
- Implemented intelligent alerting with threshold rules, deduplication, and severity-based routing.

Aircraft Systems Simulator Ongoing
C++, OOP, System Design — GitHub

- Developed modular Aircraft Systems Simulator in C++ using OOP principles for fuel, navigation, engine, and flight control subsystems.
- Designed scalable architecture with independent subsystem classes demonstrating encapsulation, abstraction, and modularity.
- Implemented real-time simulation of aircraft parameters and fault detection mechanism for system failures.
- Applied systems engineering concepts including subsystem integration, interface design, and fault management.

PROFESSIONAL EXPERIENCE

Fitoor-e-Kala — International Cultural Festival Nov 2024 – Dec 2024
Co-Lead, Tech & Events — KL University — Coordinated large-scale festival managing technology infrastructure, registration systems, and volunteer coordination.

Google Developer Groups (GDG) On Campus Sep 2025 – Present
Web Development Lead & Community Educator — Conduct workshops on cloud-native development, DevOps, containerization, and infrastructure automation.

CERTIFICATIONS & ACHIEVEMENTS

Oracle Cloud Infrastructure (OCI) Certified DevOps Professional — Cloud deployment, infrastructure automation, CI/CD.
1st Place — ZeroOne Hackathon — Systems design and rapid prototyping.